Data Processing

Introduction to ARM Cortex-M Microprocessors — Chapter 6

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Why This Chapter Matters

Every peripheral is controlled by setting and clearing single bits inside a 32-bit register, without disturbing neighbouring bits. Load, modify, store back: this is the most common pattern in embedded firmware. This chapter builds it from instructions. You will then know exactly what `GPIOA->MODER |= (1U << 10)` executes, and why.

After this chapter, you can:

1. Choose the right arithmetic instruction; explain why a 32-bit constant may need two.
2. Write a read-modify-write that spares neighbouring bits.
3. Predict shift results; tell logical from arithmetic shifts.
4. Use the flexible second operand and barrel shifter.
5. Use offset, pre-indexed and post-indexed addressing.
6. Use a scaled register offset to reach an array element.

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Arithmetic Instructions

Instruction	Operation	Note
`ADD`	`Rd = Rn + Operand2`	
`SUB`	`Rd = Rn − Operand2`	
`RSB`	`Rd = Operand2 − Rn`	Negation: `RSB R0, R1, #0`
`MUL`	`Rd = Rn × Rm`	Single cycle on the M4
`SDIV` / `UDIV`	`Rd = Rn ÷ Rm`	2–12 cycles

Loading a 32-bit constant

The immediate value must fit *inside* the instruction, which is itself only 32 bits. So a full constant needs either `MOVW` then `MOVT` (two instructions, no memory access), or `LDR Rn, =value` (one instruction plus a load from a literal pool).

Arithmetic in C and in Assembly Load–process–store, once more

In C

```
total = a + b - c;
```

In assembly

```
    LDR  R0, =a
    LDR  R0, [R0]   ; R0 = a
    LDR  R1, =b
    LDR  R1, [R1]   ; R1 = b
    LDR  R2, =c
    LDR  R2, [R2]   ; R2 = c
    ADD  R3, R0, R1 ; R3 = a + b
    SUB  R3, R3, R2 ; R3 = (a+b) - c
    LDR  R4, =total
    STR  R3, [R4]
```

One C statement, one register-only computation

Every operand is loaded into a register before either `ADD` or `SUB` can touch it. Neither instruction ever reads or writes memory directly — the result is stored only once, at the very end.

The compiler is free to reorder or reuse registers, but the shape never changes: load every operand, compute in registers, store the result.

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Logical Instructions

Instruction	Operation	What it is for
`AND`	`Rd = Rn & Op2`	**Masking** — keep only bits you care about
`ORR`	`Rd = Rn | Op2`	**Setting** bits to 1
`EOR`	`Rd = Rn ^ Op2`	**Toggling** bits
`BIC`	`Rd = Rn & ~Op2`	**Clearing** bits to 0

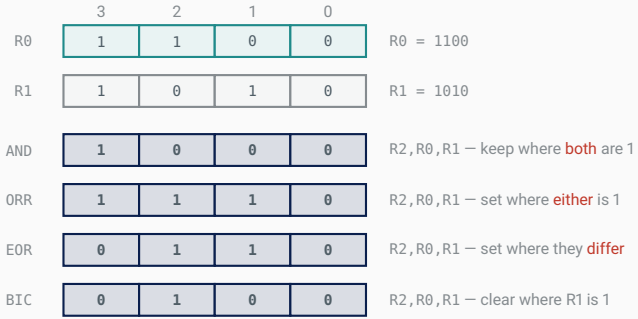
`BIC` is the one without a matching C operator

In C you clear bits with `x &= ~mask`. ARM gives this its own instruction. So the compiler emits *one* `BIC`, where you might expect `MVN` followed by `AND`.

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The Four Operations on the Same Bits

R0 contains 12_{10}, R1 contains 10_{10}.



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Read–Modify–Write The sequence behind every peripheral write

In C

```
GPIOA->MODER &=
        ~(3U << 10);

GPIOA->MODER |=
        (1U << 10);
```

In assembly

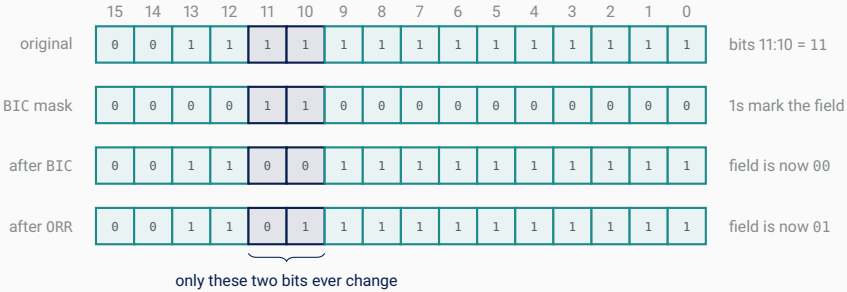
```
    LDR  R3, =0x40020000
    LDR  R0, [R3]
    BIC  R0, R0, #0xC00   ; clear bits 11:10
    ORR  R0, R0, #0x400   ; set bit 10
    STR  R0, [R3]
```

Why you must keep the neighbouring bits

This register holds settings for sixteen pins. Writing the whole register would reconfigure every one of them. Load it, change only your bits, then store it back.

`3U << 10 = 0xC00;` `1U << 10 = 0x400`. This exact sequence sets up PA5 as an output.

Following the Bits Through Clear-Then-Set



`ORR` alone cannot turn `11` into `01` — OR can only turn bits on. `BIC` must clear the field first.

Assembly Step by C Step

Step	Assembly	Equivalent C
Load the register	`LDR R1, [R0]`	implicit in `GPIOA->MODER`
Clear the field	`BIC R1, R1, R2`	`&= ~(3U << 10)`
Set the new value	`ORR R1, R1, #0x400`	`|= (1U << 10)`
Store the register	`STR R1, [R0]`	implicit in the `->` assignment

Two rows are “implicit”

C never issues a separate load or store on its own — the compiler emits `LDR`/`STR` automatically around every `->MODER` reference. Only the middle two rows have a direct C spelling.

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Read-Modify-Write at Five Levels

Level	Representation
Physical hardware	Two configuration flip-flops selecting the PA5 output driver
Register	`GPIOA_MODER` bits 11:10
C	`MODER &= ~(3U « 10); MODER |= (1U « 10);`
Assembly	`LDR` / `BIC` / `ORR` / `STR`
Machine code	Four Thumb-2 encodings

One operation, five simultaneous descriptions

Read downward: a flip-flop is a bit, a bit is in a register, a register has a name in C, a name becomes four instructions, four instructions become bytes in Flash. Read upward: the C statement reaches a flip-flop.

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The Four Shifts

Shift operations move all bits in a register left or right by a specified number of positions, useful for fast arithmetic and bit manipulation.

Connection to Digital Electronics

You have already encountered these concepts in Digital Electronics as hardware shift registers (SISO, SIPO, PISO, PIPO) and ring counters.

Instruction	Meaning	Effect
`LSL #n`	Logical shift left	Multiplies by 2^n; zeros fill in from the right
`LSR #n`	Logical shift right	Divides an unsigned value by 2^n; zeros fill in from the left
`ASR #n`	Arithmetic shift right	Divides a signed value by 2^n; the sign bit fills in from the left
`ROR #n`	Rotate right	Bits that fall off the right re-enter on the left

Only one bit decides LSR versus ASR

Both shift right. The difference is what fills the vacated bits: `LSR` always fills with 0. `ASR` copies the sign bit, so a negative number stays negative after the shift.

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Shifts as Ordinary Instructions

```
LSL R0, R1, #2    ; R0 = R1 << 2   (R1 times 4)
LSR R0, R1, #1    ; R0 = R1 >> 1   (R1 divided by 2, unsigned)
ASR R0, R1, #1    ; R0 = R1 >> 1   (R1 divided by 2, signed)
```

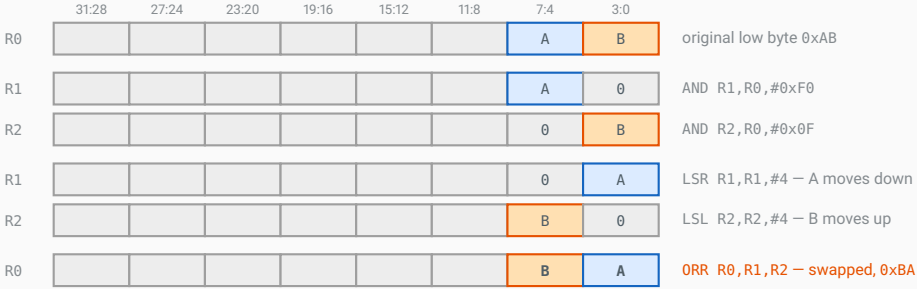
Written this way, a shift is a complete instruction: it reads `R1`, shifts it, and writes the result to `R0` — exactly like `ADD` or `MOV`.

The idea worth carrying forward

A shift is useful by itself. But ARM can also feed a shifted value directly into another instruction, without a separate shift step first. That is the subject of the next section.

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Swapping Two Nibbles Isolate with AND, recombine with ORR



Multiplying by a Power of Two Strength reduction: replacing multiply with shift

In C

```
total = 4 * a + b;
```

a and b are already in R0 and R1 — no LDR needed here.

In assembly

```
    LSL  R2, R0, #2  ; R2 = a << 2
                     ;    = a * 4
    ADD  R3, R2, R1  ; R3 = (4*a) + b
```

Why the compiler prefers LSL over MUL

Multiplying by 4 is multiplying by 2^2. A left shift by 2 bits produces exactly that result, and on most cores a shift is cheaper than a general multiply. The compiler makes this substitution automatically — it is called strength reduction. No MUL instruction appears at all.

This only works for a power-of-two constant. `4 * a` becomes a shift; `5 * a` still needs a real MUL.

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What is the Barrel Shifter?

A **barrel shifter** is a specialized digital circuit that can shift or rotate a data word by any number of bit positions in a single clock cycle.

Why is it a game-changer in ARM?

- **Hardware Efficiency:** Unlike a basic shift register (which shifts 1 bit per cycle), it uses combinational multiplexers to shift *n* bits instantly.
- **Zero-Cost Operations:** In ARM processors, it is wired directly into the ALU input path.
- **Combined Execution:** Shifts and arithmetic/logic operations happen in the same instruction cycle with zero clock penalty.

`ADD R0, R1, R2, LSL #2` ⟶ Shift `R2` left by 2, then add to `R1` in 1 cycle.

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The Flexible Second Operand

```
Opcode{S}{cond}   Rd,   Rn,   Operand2
```

Form	Written	Meaning
Constant	`#constant`	An immediate encoded in the instruction
Register	`Rm`	A register holding the second operand
Register with shift	`Rm, LSL #n`	`Rm` shifted before use

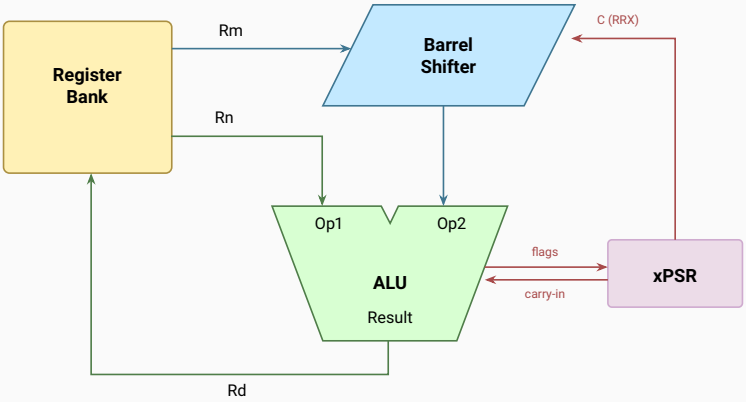
```
ADD R0, R1, R2, LSR #3    ; R0 = R1 + (R2 >> 3)
ORR R0, R1, R2, LSL #2    ; R0 = R1 | (R2 << 2)
```

Folded into the instruction, not a separate step

The barrel shifter sits in the datapath *before* the ALU, so the shift and the operation happen together: no separate shift instruction is needed. `Rm` itself stays unchanged — only the value entering the ALU is shifted.

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Where the Barrel Shifter Sits



`Rm` feeds the barrel shifter. Its output enters the ALU as `Op2`. `Rn` feeds `Op1` directly. The result is written back to `Rd`, and the flags update in xPSR. The carry bit of xPSR is sent to ALU and barrel shifter.

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The Four Addressing Modes

Assume `R1` = `0x2000` before each instruction.

Instruction	Mode	Reads from	R1 afterwards
`LDR R0, [R1]`	register	`0x2000`	`0x2000`
`LDR R0, [R1, #8]`	offset	`0x2008`	`0x2000`
`LDR R0, [R1, #8]!`	pre-indexed	`0x2008`	`0x2008`
`LDR R0, [R1], #8`	post-indexed	`0x2000`	`0x2008`

The Writeback Indicator (`!`)

- The exclamation point `!` explicitly commands auto-update (writeback) for pre-indexed access.
- Post-indexed access **implicitly updates** the base register without requiring a `!` symbol.

1. Simple offset

```
LDR R0, [R1, #8]
; address = R1 + 8
; R1 unchanged
```

2. Pre-indexed: Calculate first, then access.

```
LDR R0, [R1, #8]!
; address = R1 + 8
; R1 = R1 + 8
```

3. Post-indexed: Access first, then calculate.

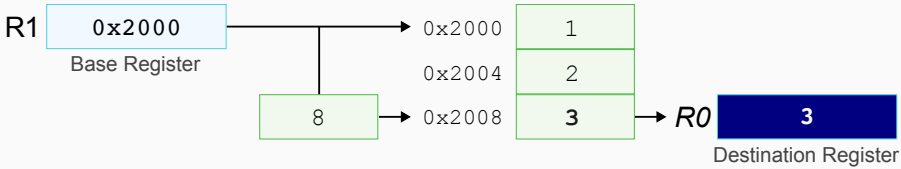
```
LDR R0, [R1], #8
; address = R1
; R1 = R1 + 8 (after)
```

Read the brackets, not the mnemonic

- Offset inside the brackets → pre-indexed.
- Offset outside the brackets → post-indexed.
- Trailing ! → writeback.

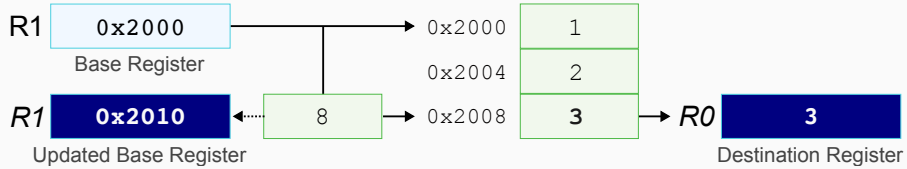
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Visualizing Offset Access `LDR R0, [R1, #8]`



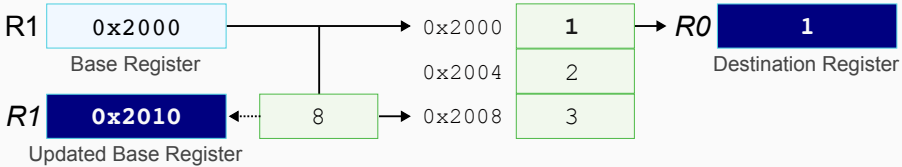
`R0` is loaded with the address pointed by `R1 + 8` — `0x2008`. `R1` is not updated.

Visualizing Pre-Indexed Access `STR R0, [R1, #16]!`



The address is computed *first*: `0x2000 + 16 = 0x2010`. This is both where `R0` is loaded from, and where `R1` ends up. One calculation, used twice.

Visualizing Post-Indexed Access `STR R0, [R1], #16`



`R0` is loaded from address `R1` already held — `0x2000` — *before* anything changes. `R1` is updated *afterward*, to `0x2010`, ready for the next access.

These Modes Are Not Just for Arrays

Offset, pre-indexed, and post-indexed addressing are useful for any pointer, not only array indices:

- A structure field at a fixed offset from a base pointer — `[R1, #4]`
- A buffer walked one element at a time, letting the pointer advance itself — `[R1], #4`
- A local variable in a stack frame at a fixed offset from the frame pointer — `[R7, #8]`

Every one of these uses a constant offset. No second register or shift is needed. The next section adds a register offset—and then a scaled register offset—when the offset itself is data, such as a run-time index.

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Dynamic Indexed Access

Form	Written	Typical C equivalent
Immediate offset	`[R1, #4]`	`*(p + 1)`
Register offset	`[R1, R2]`	`*((char *)p + offset)`
Scaled register offset	`[R1, R2, LSL #2]`	`p[i]`

The key idea

A register offset supplies a displacement at run time. A scaled register offset also multiplies that displacement by a power of two, which makes it ideal for array indexing.

For a 32-bit element:

$$\text{address} = \text{base} + (\text{index} \times 4)$$

Scaled Register Offset

```
LDR R0, [R1, R2, LSL #2]
```

- R1 — base address of the array
- R2 — the index
- LSL #2 — scale by 4, the element size

An immediate offset is fixed when you write the program; a register offset can change while the program runs, which is exactly what a loop counter needs.

Indexing without a multiply

The barrel shifter feeds directly into address generation, so scaling the index, adding the base, and loading the value all happen in one instruction. No separate shift or `MUL` is needed.

This is Chapter 3's formula, `base + (index * element size)`, computed by hardware. It is why that multiplication never appears explicitly in compiler output.

Shift	Scales by, matching
LSL #1	2 — a 16-bit halfword (`int16_t`)
LSL #2	4 — a 32-bit word (`int32_t`, any pointer)

Array Indexing Becomes a Scaled Offset

C

```
int x = array[i];
```

For a 32-bit `int`, the compiler computes the address as `array` $+ i \times 4$.

C type	Typical scaling
`int16_t *`	`LSL #1` (2 bytes)
`int32_t *`	`LSL #2` (4 bytes)
`uint32_t *`	`LSL #2` (4 bytes)

Typical ARM

```
; R1 = array
; R2 = i
LDR R0, [R1, R2, LSL #2]
```

Address calculation

$$R1 + (R2 \times 4)$$

The shift scales the index before it is added to the base address.

This is why array access on ARM is so cheap. If someone says “the barrel shifter is just a historical curiosity,” they have not looked at what the compiler produces for an array loop.

Array Indexing vs Pointer Arithmetic

Same address, different notation

```
int x = array[i];
int x = *(array + i);

int *p = array;
int x = p[i];   // same as *(p + i)
```

In every case, `i` is an *element index*. Pointer arithmetic automatically accounts for the size of the pointed-to type.

What changes at the machine level?

The compiler turns the element index into a byte displacement:

$$i \times \texttt{sizeof(int)}$$

A useful distinction

`p + i` advances by `i` *elements*.
`(char *)p + i` advances by `i` *bytes*.

Worked Example: Array Processing

C code

```c
int array[10];
int sum = 0;
int i;

for (i = 0; i < 10; i++) {
    sum += array[i];
}
```

ARM assembly

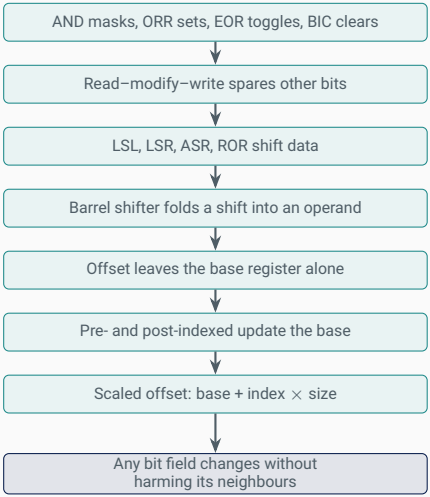
```
    LDR  R1, =array      ; base address
    MOV  R0, #0          ; sum = 0
    MOV  R2, #0          ; i = 0
loop
    LDR  R3, [R1, R2, LSL #2]
    ADD  R0, R0, R3      ; sum += a[i]
    ADD  R2, R2, #1      ; i++
    CMP  R2, #10
    BLT  loop
```

This example combines

- `LDR =` to load an address
- a scaled register offset for the index
- register-only arithmetic
- `CMP` and a conditional branch for the loop

The `LSL #2` handles the array arithmetic; no multiply instruction appears.

Summary



Terminology covered

- AND, ORR, EOR, BIC
- Shifts and rotate
- Flexible second operand
- Pre- and post-indexed
- Scaled register offset

Next: Chapter 7

Control Flow, Functions and the Stack

We can move data. Next, we decide what to do with it.

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